

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0700

North Bonneville 100 kV
February 24, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0700. The Interconnection Customer proposes a 274.6 MW wind facility interconnecting at the North Bonneville 100 kV substation in the Illinois area. The request was received on November 13, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 306 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,414,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: OXBOW - BLYTHE 100.0 kV
Contingency: BEAR CANYON - CLINTMOORE 100.0 kV
Loading: 166.4% (Rating: 352 MVA)
DFAX: 100.0%
Pre-project loading: 88.3%
2. Facility: NORTH BONNEVILLE SUBSTATION - OXBOW 100.0 kV L
Contingency: CLINTMOORE SUBSTATION - BEAR CANYON SUBSTATION 100.0 kV
Loading: 163.0% (Rating: 366 MVA)
DFAX: 100.0%
Pre-project loading: 88.0%
3. Facility: KENZIE CRK - OREGON CITY 100.0 kV
Contingency: SHORELINE - FRONT RANGE 100.0 kV
Loading: 149.8% (Rating: 125 MVA)
DFAX: 27.2%
Pre-project loading: 90.0%
4. Facility: TOCOI - OREGON CITY SUBSTATION 100.0 kV
Contingency: SHORELINE - FRONT RANGE 100.0 kV line
Loading: 144.8% (Rating: 127 MVA)
DFAX: 27.2%
Pre-project loading: 85.8%
5. Facility: SHORELINE - FLAMBEAU HYDROELECTRIC STATION 100.0 kV
Contingency: OXBOW - NORTH BONNEVILLE 100.0 kV
Loading: 143.2% (Rating: 316 MVA)
DFAX: 81.3%
Pre-project loading: 72.5%

6. Facility: TOCOI - SHAWBORO 100.0 kV
Contingency: SHORELINE - FRONT RANGE 100.0 kV
Loading: 140.1% (Rating: 128 MVA)
DFAX: 27.2%
Pre-project loading: 82.0%
7. Facility: SHORELINE - FRONT RANGE 100.0 kV
Contingency: OXBOW - NORTH BONNEVILLE 100.0 kV line
Loading: 139.4% (Rating: 336 MVA)
DFAX: 81.3%
Pre-project loading: 73.0%
8. Facility: SOCRATES 19 - FRONT RANGE 100.0 kV Line
Contingency: OXBOW - N BONNEVILLE 100.0 kV Line
Loading: 139.4% (Rating: 336 MVA)
DFAX: 81.3%
Pre-project loading: 73.0%
9. Facility: KENZIE CREEK - SOCRATES 19 100.0 kV
Contingency: SHORELINE - FRONT RANGE 100.0 kV
Loading: 138.3% (Rating: 151 MVA)
DFAX: 27.2%
Pre-project loading: 88.8%
10. Facility: BEAR CANYON - SOCRATES 19 100.0 kV
Contingency: OXBOW - N BONNEVILLE 100.0 kV
Loading: 134.3% (Rating: 432 MVA)
DFAX: 100.0%
Pre-project loading: 70.7%
11. Facility: BEAR CANYON - CLINTMOORE 100.0 kV L
Contingency: N BONNEVILLE - OXBOW 100.0 kV
Loading: 131.8% (Rating: 453 MVA)
DFAX: 100.0%
Pre-project loading: 71.2%
12. Facility: HOOSIER FAIRVIEW - BLYTHE 100.0 kV Cir. 1
Contingency: BEAR CANYON - CLINTMOORE 100.0 kV
Loading: 118.8% (Rating: 159 MVA)
DFAX: 12.1%
Pre-project loading: 97.9%
13. Facility: BLYTHE - HOOSIER FAIRVIEW 100.0 kV L
Contingency: BEAR CANYON - CLINTMOORE SUBSTATION 100.0 kV L
Loading: 118.8% (Rating: 159 MVA)
DFAX: 12.1%
Pre-project loading: 97.9%
14. Facility: BLTHE - HOOSIER FAIRVIEW 100.0 kV Ckt 2
Contingency: CLINTMOORE - BEAR CANYON SUBSTATION 100.0 kV Line
Loading: 117.6% (Rating: 160 MVA)
DFAX: 12.1%
Pre-project loading: 96.9%
15. Facility: MATTEON - ARCH SMITH 100.0 kV
Contingency: SOUTH BOSTON - BLYTHE 230.0 kV
Loading: 110.5% (Rating: 134 MVA)
DFAX: 6.7%
Pre-project loading: 96.8%
16. Facility: SPEARFISH HYDRO - ARCH SMITH 100.0 kV Ckt 1
Contingency: SPEARFISH HYDRO SUBSTATION - ARCH SMITH 100.0 kV Ckt 2

Loading: 105.5% (Rating: 142 MVA)

DFAX: 6.9%

Pre-project loading: 92.1%

17. Facility: SPEARFISH HYDRO - ARCH SMITH 100.0 kV

Contingency: SPEARFISH HYDRO - ARCH SMITH 100.0 kV Circuit 1

Loading: 105.5% (Rating: 142 MVA)

DFAX: 6.9%

Pre-project loading: 92.1%

18. Facility: ELWOOD SUBSTATION - LINCON 230.0 kV line

Contingency: S BOSTON - BLTHE 230.0 kV

Loading: 104.0% (Rating: 511 MVA)

DFAX: 12.9%

Pre-project loading: 97.1%

19. Facility: SPEARFISH HYDRO - HOOSIER FAIRVIEW 100.0 kV Line 1

Contingency: SPEARFISH HYDRO - HOOSIER FAIRVIEW 100.0 kV #2

Loading: 101.7% (Rating: 182 MVA)

DFAX: 7.5%

Pre-project loading: 90.4%

20. Facility: HOOSIER FAIRVIEW - SPEARFISH HYDRO 100.0 kV Ckt 2

Contingency: SPEARFISH HYDRO - HOOSIER FAIRVIEW 100.0 kV Cir. 1

Loading: 101.7% (Rating: 182 MVA)

DFAX: 7.5%

Pre-project loading: 90.4%

21. Facility: SUGAR HILL - RITTER PARK 345.0 kV

Contingency: HILLTOP - PINEY CRK PROJECT 500.0 kV L

Loading: 101.1% (Rating: 1,622 MVA)

DFAX: 10.1%

Pre-project loading: 99.4%

22. Facility: S BOSTON - BLYTHE 230.0 kV

Contingency: LINCOLN - ELWOOD 230.0 kV

Loading: 100.8% (Rating: 615 MVA)

DFAX: 17.2%

Pre-project loading: 93.2%

23. Facility: LEOTA - HOOSIER FAIRVIEW 100.0 kV

Contingency: BEAR CANYON - CLINTMOORE 100.0 kV

Loading: 100.3% (Rating: 114 MVA)

DFAX: 13.1%

Pre-project loading: 68.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
OXBOW - BLYTHE 100.0 kV	Rebuild line with higher-capacity conductor	\$1,852,000	42% (\$775,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
OXBOW - N BONNEVILLE 100.0 kV	Rebuild line with higher-capacity conductor	\$16,940,000	42% (\$7,087,000)
OREGON CITY - KENZIE CREEK 100.0 kV	Rebuild line with higher-capacity conductor	\$7,466,000	32% (\$2,402,000)
TOCOI - OREGON CITY 100.0 kV Line	Rebuild line with higher-capacity conductor	\$5,577,000	32% (\$1,794,000)
SHORELINE - FLAMBEAU HYDROELECTRIC STATION 100.0 kV	Rebuild line with higher-capacity conductor	\$774,000	42% (\$324,000)
TOCOI - SHAWBORO 100.0 kV	Rebuild line with higher-capacity conductor	\$4,155,000	32% (\$1,337,000)
SHORELINE - FRONT RANGE 100.0 kV	Rebuild line with higher-capacity conductor	\$877,000	42% (\$367,000)
SOCRATES 19 - FRONT RANGE 100.0 kV Line	Rebuild line with higher-capacity conductor	\$1,969,000	42% (\$824,000)
SOCRATES 19 - KENZIE CRK 100.0 kV	Rebuild line with higher-capacity conductor	\$1,317,000	32% (\$424,000)
BEAR CANYON - SOCRATES 19 100.0 kV	Rebuild line with higher-capacity conductor	\$1,694,000	42% (\$709,000)
BEAR CANYON - CLINTMOORE 100.0 kV line	Rebuild line with higher-capacity conductor	\$3,142,000	42% (\$1,315,000)
HOOSIER FAIRVIEW - BLYTHE 100.0 kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,088,000	29% (\$615,000)
BLYTHE - HOOSIER FAIRVIEW 100.0 kV L	Rebuild line with higher-capacity conductor	\$2,993,000	29% (\$881,000)
HOOSIER FAIRVIEW - BLYTHE 100.0 kV Ckt 2	Rebuild line with higher-capacity conductor	\$2,062,000	29% (\$607,000)
ARCH SMITH - MATTESON 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,051,000	32% (\$657,000)
SPEARFISH HYDRO - ARCH SMITH 100.0 kV Ckt 1	Reconductor line	\$588,000	100% (\$588,000)
SPEARFISH HYDRO - ARCH SMITH 100.0 kV	Reconductor line	\$1,696,000	100% (\$1,696,000)
ELWOOD SUBSTATION - LINCON 230.0 kV line	Reconductor line	\$7,168,000	12% (\$860,000)
HOOSIER FIRVIEW - SPEARFISH HYDRO SUBSTATION 100.0 kV	Reconductor line	\$700,000	51% (\$356,000)
HOOSIER FAIRVIEW - SPEARFISH HYDRO 100.0 kV Ckt 2	Reconductor line	\$1,104,000	51% (\$562,000)
SUGAR HILL - RITTER PARK 345.0 kV	Reconductor line	\$6,836,000	2% (\$161,000)
SOUTH BOSTON - BLYTHE SUBSTATION 230.0 kV	Reconductor line	\$13,183,000	13% (\$1,717,000)
LEOTA - HOOSIER FAIRVIEW 100.0 kV	Reconductor line	\$1,245,000	29% (\$356,000)
	Total allocated to project		\$26,414,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.